

Downloads

Model and related tools

For questions about the code, please make sure to check the FAQs before contacting us. The SMPL model is available for non-commercial research, education, and artistic purposes only. By downloading the model you agree to the [license terms](#).

The model is also available for commercial purposes by contacting sales@meshcapade.com.

If you are unsure whether your use counts as non-commercial please contact smpl@tuebingen.mpg.de before downloading the model.

Overview:

SMPL is an easy-to-use, realistic, model of the of the human body that is useful for animation and computer vision.

SMPL lets you create realistic human body shapes with realistic deformations.

Available here is a version of SMPL that is similar to that described in [this paper](#).

SMPL for Python Users

Included in this download:

- Model files
 - PKL file format
 - separate male and female model files



- Code
 - Functions to load / save SMPL models
 - 'Hello World' sample scripts

System Requirements

Operating system: macOS, Linux

Python 2.7 Dependencies

- [Numpy & Scipy](#)
- [Chumpy](#)
- [OpenCV](#)

Download

[Download version 1.0.0 for Python 2.7 \(female/male, 10 shape PCs\)](#)

[Download version 1.1.0 for Python 2.7 \(female/male/neutral, 300 shape PCs\)](#)

[Download UV map in OBJ format](#)

Julia Models

Male (43 MB) Female (43 MB) Neutral (43 MB)

DMPLs for AMASS

- 8 DMPL components to control soft tissue dynamics
- Available for male, female, and neutral models
- Includes .npz numpy arrays
- For an example usage refer to project [AMASS](#)
- If you use DMPLs, in addition to SMPL, please also cite the reference for [Dyna](#)

Download



Download DMPLs compatible with SMPL

Sample Animated Bodies

These are FBX files that you can load and play in most graphics software



Download sample animated bodies FBX files

SMPL for Maya

Included in this download:

- Male and female models (FBX format)
 - Female model:
basicModel_f_lbs_10_207_0_v1.0.2.fbx
 - MD5:
81a57177f8cbf584f8d6379742c99e2e
 - Also referred to as:
SMPL_female_010_207.fbx
 - Male model:
basicModel_m_lbs_10_207_0_v1.0.2.fbx
 - MD5:
67542f27aa4bc6b8d12da28fdce51539
 - Also referred to as:
SMPL_male_010_207.fbx
 - Default mesh. The average SMPL body.
 - Rugged skeleton, using LBS
 - 10 shape blendshapes



- 207 corrective pose blendshapes
- UV map
- Script and data to use SMPL in Maya
 - If you want to edit body shape or repose the mesh, you need to run the script.
- README.txt for installation and license instructions

Dependencies

Numpy is required for running this script. Numpy is a Python module that can be installed following the instructions given [here](#) or [here](#)

Please make sure you have numpy installed on your computer and accessible through Maya's Python. We are working towards removing this dependency.

Download

[Release notes for version 1.0.2](#)

[Download version 1.0.2 for Maya](#)

[Download SMPL-Maya loader by BodyLabs](#)

Tutorial videos

Tutorial videos for Maya can be found in the [Videos](#) section.

SMPL for Unity

Included in this download:

- Male and female models (Unity-compatible FBX format)
 - Default mesh. The average SMPL body.
 - Rigged skeleton, using LBS



- 10 shape blendshapes
- 207 corrective pose blendshapes
- UV map
- Scripts and sample Unity project to help you get started
 - If you want to edit body shape or repose the mesh, you need the scripts.
- README.txt for installation and license instructions
- How-to_SMPLinUnity.pdf for step-by-step guide for using SMPL in Unity

System Requirements

Unity 5+ (tested on macOS Sierra and Windows 10)

Download

[Download version 1.0.0 for Unity](#)

How to use SMPL in Unity

[Download tutorial \(PDF\)](#)

SMPL for Blender

This Blender add-on allows you to add SMPL skinned meshes to your current Blender scene. Each imported SMPL mesh consists of a shape specific rig, as well as shape keys (blend shapes) for shape and pose correctives.

- Requirements: Blender 2.80+
- Additional dependencies: None
- Used SMPL model: SMPL v1.0 with 10 shape components

Installation instructions

Download SMPL Blender Add-on (20210611, 34MB)



SMPL for Unreal Plugin (USMPL)

The USMPL plugin allows a user to put the SMPL model in Unreal Engine and control its pose using ROS in real-time. The plugin is available at <https://github.com/robot-perception-group/USMPL>. Following are the gender-specific models for the USMPL plugin.

basicModel_f_lbs_10_207_0_v1.0.0.npz

basicmodel_m_lbs_10_207_0_v1.0.0.npz

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